Into the Matrix

Exploration of Molecular Symmetry and Bonding Using Jupyter Notebooks

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Trial 1

```
import numpy as np


water = np.array([
  ['atom', 'x',  'y',  'z'],
  ['O',     0,    0,     0],
  ['H',     0,    1,    -1],
  ['H',     0,   -1,    -1],
])
```

Trial 1

```
rot_x = np.array([
    [1,  0,  0],
    [0, -1,  0],
    [0,  0, -1],
])

rot_y = np.array([
    [-1,  0,  0],
    [ 0,  1,  0],
    [ 0,  0, -1],
])
```

Trial 1

```
rot_z = np.array([
    [-1,  0, 0],
    [ 0, -1, 0],
    [ 0,  0, 1],
])

ref_xy = np.array([
    [1, 0,  0],
    [0, 1,  0],
    [0, 0, -1],
])
```

Trial 2

```
%run py/water.py

water = array(
        [['atom', 'x', 'y', 'z'],
         ['O', '0', '0', '0'],
         ['H', '0', '1', '-1'],
         ['H', '0', '-1', '-1']], dtype = '<U21')
water_xyz = array(
        [[ 0,  0,  0],
         [ 0,  1, -1],
         [ 0, -1, -1]])
rot_z = array(
```

Trial 2

```
        [[-1,  0,  0],
         [ 0, -1,  0],
         [ 0,  0,  1]])
water_xyz_rot = array(
        [[ 0,  0,  0],
         [ 0,  1,  1],
         [ 0, -1,  1]])
water_xyz_ref = array(
        [[ 0,  0,  0],
         [ 0,  1,  1],
         [ 0, -1,  1]])
```